



Department of Information Technology
Academic Year 2023 - 2024 (Even Semester)

Degree, Semester & Branch: IV Semester B.Tech. IT

Course Code & Title: CS3452 & Theory of Computation

Name of the Faculty member (s): Dr.V. Anusuya, ASCP/IT

Innovative Practice Description

- **Unit / Topic:** Unit IV / Normal forms for CFG, Simplification of CFG
- **Course Outcome:** CO4
- **Topic Learning Outcome:** TLO12
- **Activity Chosen:** Mind map
- **Justification:**

In unit IV, Normal forms for CFG, Simplification of CFG is an important topic asked in most of the university questions. This activity helps the students to recall the different method for simplification of CFG.

- **Time Allotted for the Activity:** 5 Minutes
- **Details of the Implementation:**
 - I explained the topic Normal forms for CFG, Simplification of CFG in the previous days.
 - The time allotted for the topic is 5 Minutes.
 - The students were asked to study the concepts of Elimination of epsilon production, Elimination of unit production, Removal of useless symbols.
 - Next day asked the students to recall the concept and ask them to draw a mind map.
 - The students completed the activity, the sample of activity as shown in Figure 1,2

• **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO8	PO9	PO10	PO12	PSO1	PSO2	PSO3
CO4	2	3	3	2	2	1	2	2	2	2	2

(1 – Low 2 – Moderate 3 – High)

• **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO4	PO8	PO9	PO10	PO12	PSO1	PSO2	PSO3
	2	3	3	2	2	1	2	2	2	2	2
Justification for correlation	Apply basic Knowledge and fundamentals of Normal forms for CFG, Simplification of CFG	The students will be able to identify the need for equivalence of Normal forms for CFG, Simplification of CFG for solving complex engineering problems	Able to design solutions for any applications using Normal forms for CFG, Simplification of CFG	Able to analyze and apply equivalence of Normal forms for CFG, Simplification of CFG	Able to follow ethics for solving the problem	Able to implement the problem individually	Able to implement equivalence of Normal forms for CFG, Simplification of CFG	Ability to recognize the need for Normal forms for CFG, Simplification of CFG in real world applications	Exhibit knowledge in Normal forms for CFG, Simplification of CFG for application development	Exhibit knowledge in Normal forms for CFG, Simplification of CFG in TOC of IoT, Cloud Computing and Data Analytics to provide IT solutions	Exhibit knowledge in Normal forms for CFG, Simplification of CFG in software projects to deliver a quality product.

- Images / Screenshot of the practice:

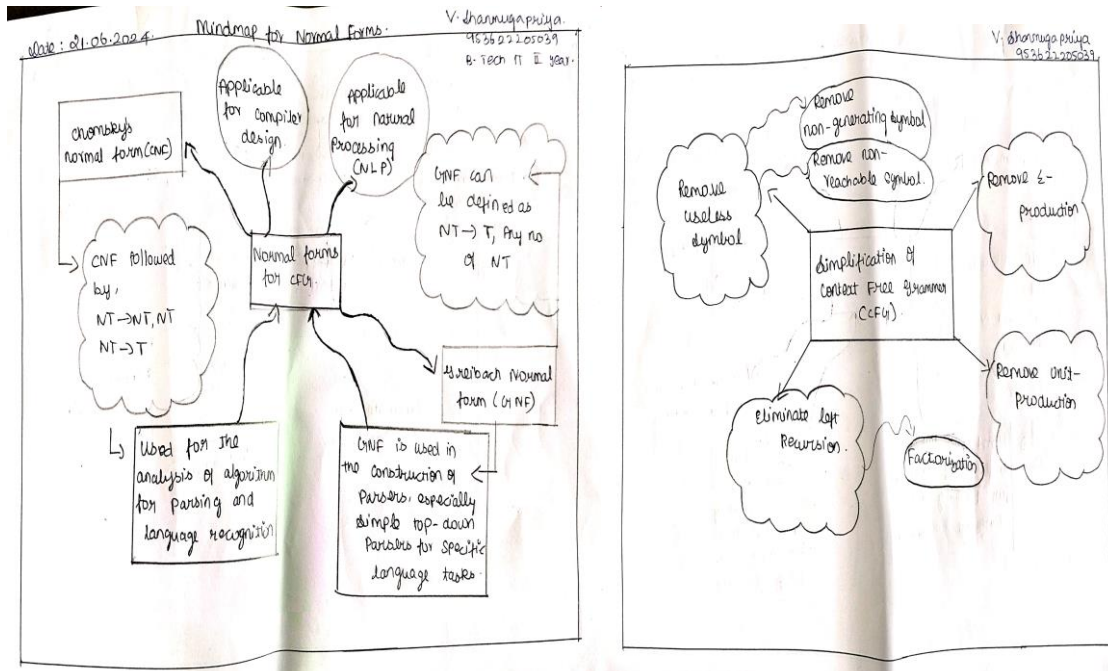


Figure 1: Mindmap of V. Shanmugapriya Simplifications of CFG

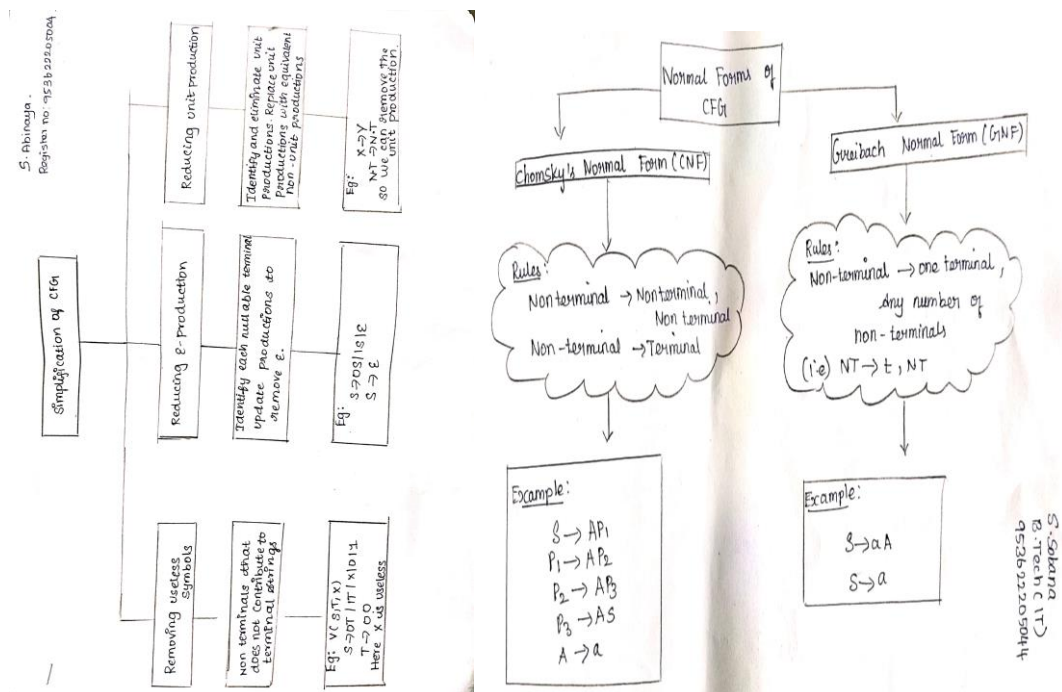


Figure 2: Mind map Simplifications of CFG of Abinaya S and Sobana S

- Reflective Critique:

- ❖ Feedback of practice from students and other stakeholders:

- Students reported that they could recall and retain the ideas of CFG simplification and normal forms.

- The students reported that the activity helped them remember everything that was covered in the previous class and that it enhanced their ability to learn on their own.
- It teaches students about the simplicity of CFG.

❖ ***Benefit of the practice:***

- It enhances students' problem-solving abilities and helps them remember how to eliminate unnecessary symbols, eliminate epsilon production, and eliminate unit production.
- The students can place their own views in the mind map

❖ ***Challenges faced in implementation:***

- Some students found it challenging to formulate mind map and several of the questions were hard to understand which elimination we have to use first.
- I thought there was a lack of clarity on the slow learners. I gave each student a distinct explanation, and after that, they understood perfectly.

References:

- ❖ <https://ohioline.osu.edu/>
- ❖ <https://www.ritrjpm.ac.in/images/computer-science/>
- ❖ <https://equiip.eu/activity/activity-5-mind-mapping/>

Signature of Faculty Member

HOD